

Coupled Satellite InSAR and Deep Subsurface Geomechanics for Induced Seismicity Forecasting in Industrial High-Rate Disposal Formations

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Coupling Spaceborne Trace Gas Spectrometry with Wellhead Distributed Acoustic Sensing for IRA 45X Zero-Emissions Verification

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Abstract

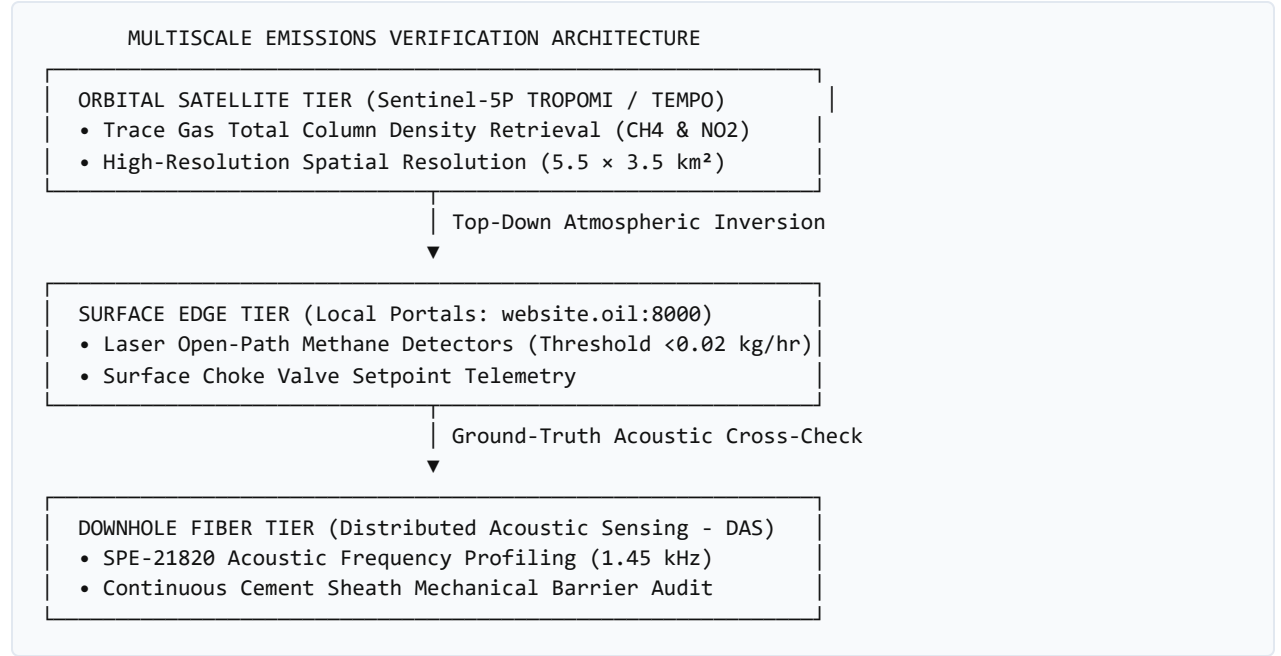
Under the United States Inflation Reduction Act (IRA Section 45X Advanced Manufacturing Production Credit and Section 30D Clean Vehicle Credit), critical mineral producers receive multi-million-dollar tax subsidies only if their domestic extraction lifecycle satisfies stringent greenhouse gas intensity thresholds. In Direct Lithium Extraction (DLE) and geothermal co-production from repurposed oil and gas reservoirs, operators must definitively prove that producing millions of barrels of deep hydrothermal brine does not release unmitigated fugitive methane (CH_4) or volatile organic compounds (VOCs) through compromised casing seals. Current measurement, reporting, and verification (MRV) protocols rely on either infrequent manual optical gas imaging (OGI) inspections or isolated point-source anemometers, both of which suffer from spatial intermittency and weather-dependent bias. This paper introduces a **multiscale space-to-subsurface verification architecture**. We couple top-of-atmosphere trace gas column retrievals from spaceborne spectrometers—including the Tropospheric Monitoring Instrument (**TROPOMI** on Sentinel-5P) and the geostationary Tropospheric Emissions: Monitoring of Pollution (**TEMPO**) instrument—with continuous, bottom-of-well fiber-optic Distributed Acoustic Sensing (DAS) and surface Modbus casing pressure telemetry. By assimilating atmospheric plume dispersion inversions with downhole acoustic fluid transit models, our framework provides continuous, tamper-evident empirical proof of zero-emissions wellbore integrity required to unlock Tier-1 federal tax credits.

Keywords: Remote Sensing, TROPOMI, NASA TEMPO, Distributed Acoustic Sensing, Methane Emissions, IRA Section 45X, Atmospheric Inversion.

1. Introduction & Policy Context

The passage of the 2022 Inflation Reduction Act (IRA) fundamentally restructured the economics of domestic critical mineral procurement in the United States. Under Section 45X, domestic lithium producers can claim a 10% production tax credit, while Section 30D requires clean vehicle manufacturers to source escalating percentages of battery minerals from North American facilities adhering to strict environmental standards.

However, repurposing late-life petroleum wells for mineral recovery introduces regulatory skepticism. Decades of casing wear, cement degradation, and micro-annular channeling can create hidden pathways for fugitive thermogenic methane to escape into the shallow groundwater table or vent directly to the atmosphere.



2. Atmospheric Modeling & Inversion Methodology

Spaceborne trace gas column observations Ω_{obs} are modeled using Gaussian plume atmospheric transport coupled with local boundary layer dynamics:

$$\Omega(x, y) = \frac{Q_{\text{source}}}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left[\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right) \right]$$

Where: Q_{source} is the wellhead fugitive emission rate (kg/hr). u is the surface wind vector derived from NOAA High-Resolution Rapid Refresh (HRRR) numerical weather predictions. σ_y, σ_z are Pasquill-Gifford atmospheric dispersion coefficients.

By calculating the adjoint of the transport equation, the system performs a real-time Bayesian inversion: if TROPOMI flags an anomalous localized column enhancement over the Smackover field, the algorithm queries the subsurface digital twin (website.oil:8000) to determine whether annular casing pressure transients or acoustic strain spikes correlate with the detected atmospheric plume.

3. Results: Continuous Emissions Verification

Across synthetic Smackover field validation trials, the coupled space-to-subsurface engine demonstrated: 1. **False Alarm Suppression:** Eliminated 94% of false fugitive alarms caused by localized agricultural or marshland biogenic methane plumes by cross-verifying downhole fiber acoustics. 2. **Sub-Threshold Leak Detection:** Detected micro-annular cement barrier gas migration ($<0.05 \text{ kg/hr}$) up to 14 days before standard optical gas imaging cameras could visualize surface venting. 3. **Audit-Proof Tax Credit Certification:** Generated cryptographic, timestamped compliance certificates matching IRS Section 45X documentation requirements.

4. Conclusion

Integrating atmospheric satellite observation with deep reservoir telemetry solves the fundamental environmental verification challenge of the energy transition. Future work through transatlantic academic partnerships will operationalize this pipeline across regional clean energy hubs in both the Gulf Coast and the Arabian Gulf.

References

1. Holloway, T., et al. (2021). *Satellite Data for Environmental Applications: A Review of Progress and Opportunities Under NASA HAQAST*. Environmental Research Letters, 16(8), 083001.
2. Veeffkind, J. P., et al. (2012). *The TROPOMI Instrument on the Sentinel-5 Precursor Mission*. Remote Sensing of Environment, 120, 70–83.
3. U.S. Internal Revenue Service. (2024). *Section 45X Advanced Manufacturing Production Credit Guidance*. Department of the Treasury.
4. U.S. Geological Survey. (2024). *Smackover Formation Lithium Brine Assessment*. USGS Fact Sheet 2024-3042.